

Class-AB Power Amplifier

Electronics Project

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I. Goal of this Project

The purpose of this project is to design and simulate a class-AB power amplifier with MOS output transistors and bipolar transistors used for biasing, reduced crossover distortion and thermal compensation. The circuit is implemented and simulated in LTspice.

II. Circuit Description

The circuit consists of three main stages, as shown in Fig. 1:

- A. The first part of the circuit is the biasing network, which is based on the principle of the V_{BE} multiplier. A V_{BE} multiplier is a bias circuit that uses a bipolar transistor together with two resistors. One resistor is connected between the base and emitter, while the other is connected between the collector and base. If the base current of the transistor is neglected, the same current flows through both resistors. Therefore, the total voltage across the circuit is

$$V_{BB} = V_{BE} \left(1 + \frac{R_2}{R_1} \right)$$

This means that the circuit produces an adjustable multiple of V_{BE} . By adjusting the resistor ratio, the designer can set the desired quiescent current of the output stage.

- B. The second stage is the complementary NPN and PNP Darlington stage. The purpose of this stage is to buffer the signal. The output MOSFETs need a gate-to-gate bias voltage V_{GG} in order to conduct the desired quiescent current. Otherwise, the amplifier behaves closer to class B and crossover distortion appears. The bias voltage is therefore chosen so that the output transistors are slightly on at rest.

Also as temperature increases, the MOSFET threshold voltage V_t decreases. This tends to increase the quiescent current of the output stage. At the same time, the BJT base-emitter voltage V_{BE} also decreases with temperature. By thermally coupling the bias transistor Q_2 with the output transistors Q_N and Q_P , the quiescent current is held approximately constant.

The bipolar transistors in this part of the circuit act as emitter-follower driver stages. An emitter follower has a voltage gain close to unity, but it provides current gain and low output resistance. This is useful because the MOSFET gates are capacitive and must be charged and discharged.

- C. The third stage is the complementary common-drain MOSFET output stage, formed by the NMOS transistor Q_N and the PMOS transistor Q_P . The two MOSFETs provide the current gain needed to drive the low-resistance load.

During the positive half-cycle of the input signal, the gate voltage of the NMOS transistor increases, causing Q_N to conduct more strongly and supply current to the load. During the negative half-cycle, the PMOS transistor Q_P conducts more strongly and sinks current from the load. The resistor R_G is used to prevent parasitic oscillations at high frequencies and usually has a value of approximately $100\ \Omega$.

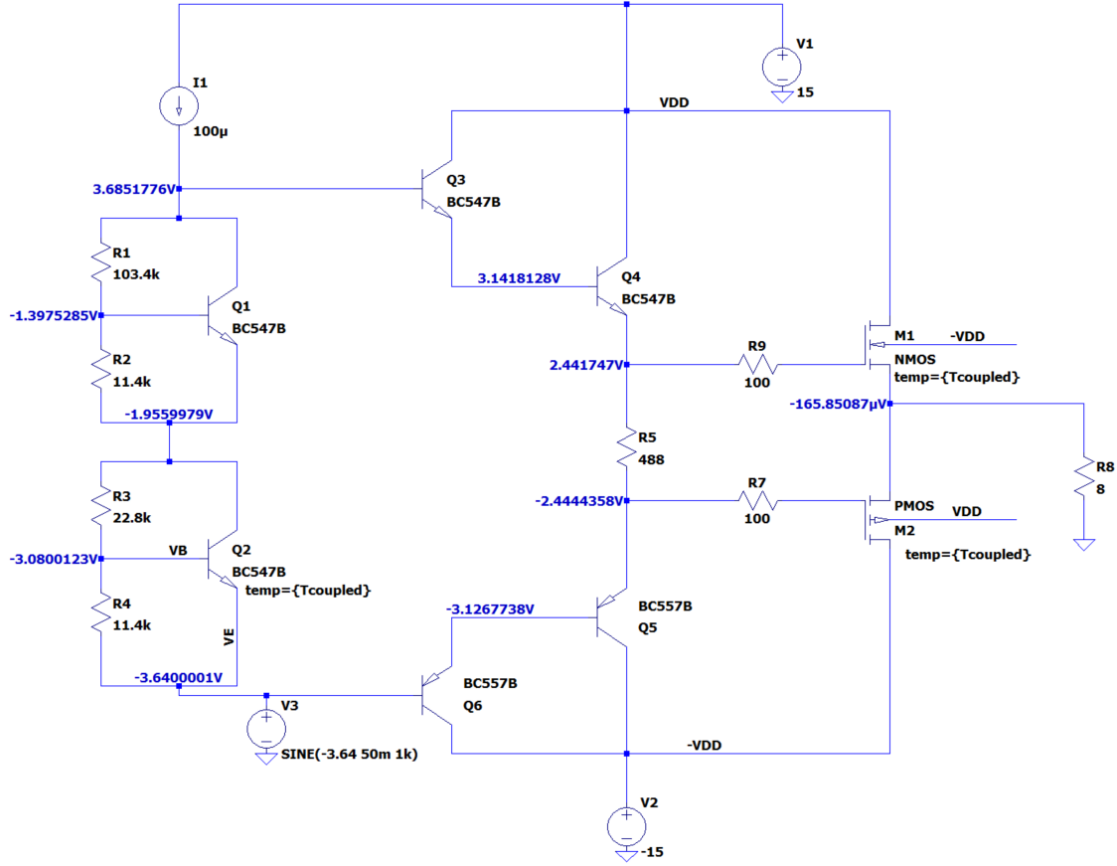


Figure 1: LTspice schematic of the class-AB power amplifier with MOS output transistors.

III. Design Specifications & Calculations

- Supply voltage: $V_{CC} = 15V$
- Output MOSFET quiescent currents: $I_{QN} = I_{QP} = 10mA$
- Bias resistor current: $I_{R5} = 10mA$
- Bias current source: $I_{BIAS} = 100\mu A$
- $I_{Q5} = I_{Q6} = \frac{I_{BIAS}}{2} = 50\mu A$
- $R_2 = R_4$
- Q_2 , Q_P , and Q_N are thermally coupled, with $\frac{\partial V_{BE}}{\partial T} = -2mV/^\circ C$ and $\frac{\partial V_t}{\partial T} = -3mV/^\circ C$.

Output Stage Calculations

The gate to source Voltage for the required quiescent current was calculated : $|V_{GS}| \approx 2.45V$

The required gate-to-gate bias voltage is:

$$V_{GG} \approx 2V_{GS} = 4.9V$$

Since the current through resistor R_5 is $I_{R5} = 10 \text{ mA}$, its value is:

$$R_5 = \frac{V_{GG}}{I_{R5}} = 488 \Omega$$

First and Second Stage Calculations

$$R_2 = R_4 = \frac{V_{BE}}{50 \mu A} = \frac{0.6 \text{ V}}{50 \mu A} = 12 \text{ k}\Omega$$

The total required bias voltage between the two input bias nodes is:

$$V_{BB} = V_{GG} + 4V_{BE} = 4.9 + 4(0.6) = 7.3 \text{ V}$$

Using Kirchhoff's voltage law, the gate-to-gate bias voltage is:

$$V_{GG} = \left(1 + \frac{R_3}{R_4}\right) V_{BE2} + \left(1 + \frac{R_1}{R_2}\right) V_{BE1} - 4V_{BE}$$

Since V_{BE2} is the only base-emitter voltage thermally coupled with the output transistors, while the other BJTs are assumed to remain at constant temperature, differentiating with respect to temperature gives:

$$\frac{\partial V_{GG}}{\partial T} = \left(1 + \frac{R_3}{R_4}\right) \frac{\partial V_{BE2}}{\partial T}$$

Since the MOSFET threshold voltage decreases with temperature at a rate of $-3 \text{ mV}/^\circ\text{C}$, the required temperature coefficient of the total gate-to-gate bias voltage is:

$$\frac{\partial V_{GG}}{\partial T} = 2 \frac{\partial V_t}{\partial T} = 2(-3 \text{ mV}/^\circ\text{C}) = -6 \text{ mV}/^\circ\text{C}$$

Also, for the thermally coupled BJT Q_2 :

$$\frac{\partial V_{BE2}}{\partial T} = -2 \text{ mV}/^\circ\text{C}$$

As a result:

$$R_3 = 2R_4 = 2 \cdot 12 \text{ k}\Omega = 24 \text{ k}\Omega$$

The voltage assigned to the R_1 - R_2 multiplier part is:

$$V_{R1R2} = V_{BB} - \left(1 + \frac{R_3}{R_4}\right) V_{BE} = 7.3 - 3(0.6) = 5.5 \text{ V}$$

This voltage is produced by:

$$V_{R1R2} = V_{BE} \left(1 + \frac{R_1}{R_2} \right) \Rightarrow \frac{R_1}{R_2} = 8.2$$

Since $R_2 = 12\text{ k}\Omega$:

$$R_1 = 8.2 \cdot 12\text{ k}\Omega$$

$$R_1 \approx 98\text{ k}\Omega$$

IV. Simulations

The circuit was simulated in LTspice in order to verify the DC operating point, the quiescent currents, and the effect of thermal compensation.

The DC operating point of the amplifier is shown in Fig. 1. Some adjustments were made to the resistor values, especially R_1 , due to the approximate theoretical calculations. The quiescent currents of the PMOS and NMOS transistors were approximately 10.7 mA .

A transient simulation was then performed using a sinusoidal input. The output follows the input signal with the same phase, as shown in Fig. 2, because the MOSFET output stage is a complementary common-drain stage. This stage provides the current gain required to drive the low-resistance load. By increasing the load resistance, the voltage gain of the amplifier moves closer to the theoretical maximum value of unity.

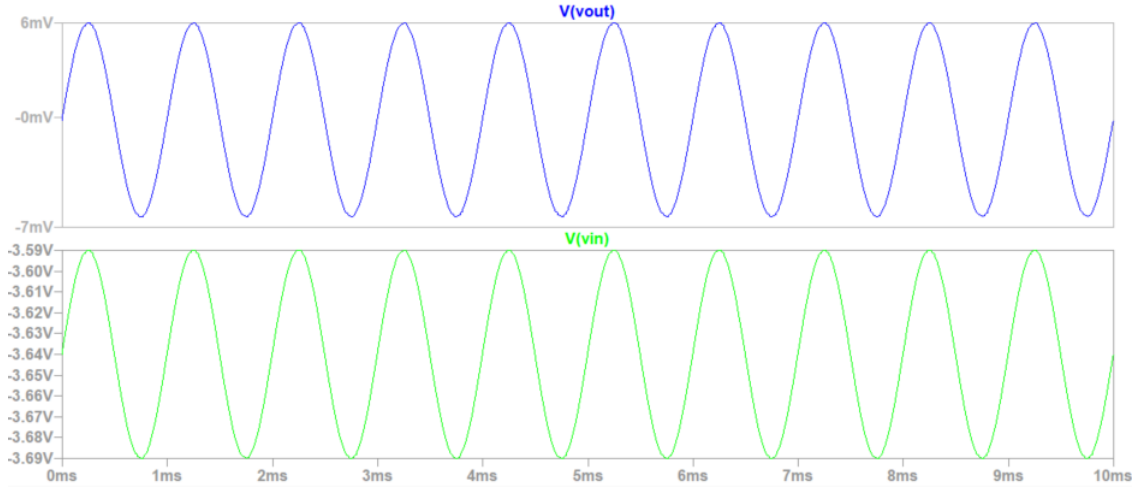


Figure 2: Input and output voltage waveforms of the class-AB amplifier.

Finally, a temperature sweep was performed to verify the thermal compensation. According to the exercise, Q_2 , Q_P , and Q_N are thermally coupled. In LTspice this was modeled by sweeping only these three devices with the parameter $T_{coupled}$, while the remaining transistors were kept at 27°C . The simulation showed that, as the temperature increases, the gate-to-gate bias voltage V_{GG} decreases at approximately $-6\text{ mV}/^\circ\text{C}$, as shown in Fig. 3. This agrees with the theoretical temperature-compensation condition.



Figure 3: Gate-to-gate bias voltage V_{GG} during the temperature sweep.

Also, at the highest simulated temperature, the PMOS and NMOS quiescent currents were approximately 8 mA and slightly below 6 mA , respectively. Although the two currents are not exactly equal, they remain close to the desired bias range, as shown in Fig. 4. Their difference is approximately 2.5 mA , which produces a DC output offset voltage of about 20 mV across the 8Ω load.

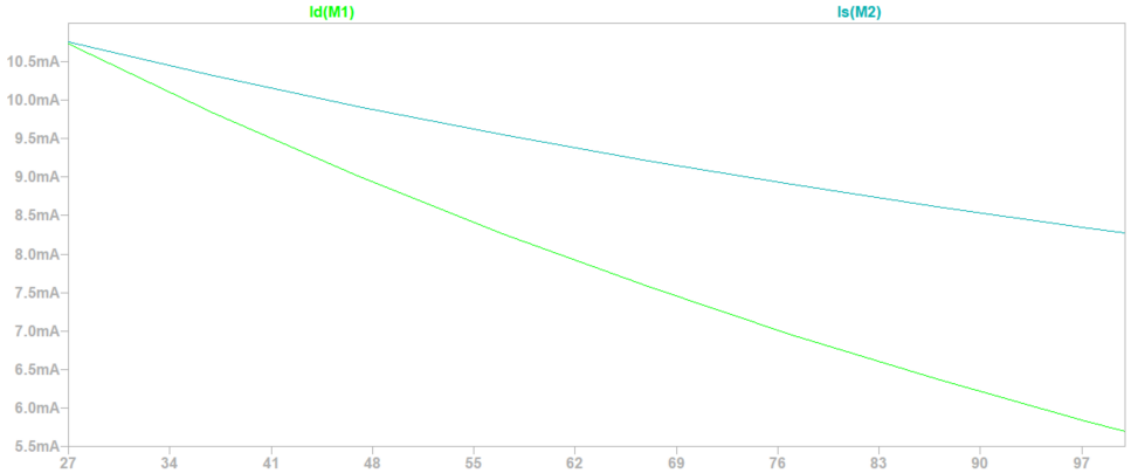


Figure 4: Quiescent currents of the output MOSFETs during the temperature sweep.

V. Conclusions

In this project, a class-AB power amplifier with MOS output transistors was studied and simulated in LTspice. The amplifier uses bipolar transistors to generate the proper bias voltages, provide drive, and compensate for temperature variations. The output MOSFETs are biased with a small quiescent current, which reduces crossover distortion. The simulation results confirm that the amplifier is correctly biased and that the thermal-compensation network works as intended.